



Nicholas Funari Voltani <nicholasvoltani@gmail.com>

About our conversation in ICTP-Unesp and Category Theory

5 mensagens

Nicholas Funari Voltani <nicholasvoltani@gmail.com>

5 de novembro de 2023 às 19:28

Para: "hgsolari@gmail.com" <hgsolari@gmail.com>

Good afternoon, prof. Hope you're doing well. Sending an email (also) to keep in touch.

As we talked on Friday, in the last months I've been interested in learning more about Category Theory, in particular Topos Theory, motivated mostly by a very interesting paper called "[What is a Thing?](#)", which is an interesting approach to Foundations of Physics using Category Theory/Topos Theory. The main idea which motivates this approach is the very close relationship between Topoi and Logic — not only Propositional Logic (which Classical Physics naturally inherits), but also Intuitionistic Logic/Heyting Algebras (which Quantum Physics appears to inherit, didn't read that much into this paper).

Category Theory gets a bad-rep of being called "Abstract Nonsense", but recently there's been a growth in so-called "Applied Category Theory" (one of its most known promoters is John Baez, who's a Mathematical Physicist). I wouldn't really want to work with pure Mathematics, so ACT seems to be a very interesting and growing field, related to many fields as far as Philosophy, [Epidemiology \(in which John Baez's been involved recently\)](#), etc... which doesn't yet have much activity in Brazil, as far as I know.

I'd really appreciate if you have any recommendations related to the above (possible graduate programs, professors who'd work on it, etc), any directions whatsoever, and also if you're currently accepting any graduate students.

Hernán Gustavo Solari <hgsolari@gmail.com>

6 de novembro de 2023 às 00:37

Para: Nicholas Funari Voltani <nicholasvoltani@gmail.com>

Dear Nicholas

thank you for writing. I tried to browse through the references you sent. The paper "What is a thing?" took me by surprise. It is fine with me to quote a sentence or a paragraph by Heidegger. Heidegger is a cryptic author for me, which means I have little to no affinity with his thought, although I recognize a few paragraphs which deserve to be read. The authors of the paper introduce wording based in some of Heidegger's more cherished metaphysical ideas as the "dasein". The authors coined: "dasenisation". My alarm goes on. A philosopher lives by his ideas, so his life reflects his ideas. Heidegger was a Nazi party member and his philosophy needs to be in correspondence with his life. Although he later put words repenting of his acts, the words do no cure the damage. For those things that have no remedy the only solution is not doing them, there is no come back from such mistakes.

does physics requires metaphysics? the limits between physics and metaphysics are not clear, but we are certain than there is no need to enter metaphysics in depth to understand physics and it will be an error so doing.

I then skipped the abstract and went to the introduction. It is also very frustrating. Have these authors ever understood something by themselves? They certainly do not understand how theories relate to the physika (natural things in greek). They believe, as most physicist do, that the unifying point is reached some how going deeper in the directions taken. This is always a certainty for the dogmatic people irrespective of the dogma held. Any critical philosopher knows that the point of excision, the branching point, is behind us. We have made mistakes and unless we cure them there is no form of producing a unified view. The mistakes are gross and evident but our formation is one proper of indoctrination. Try to explain that God is not needed to a member of any religious community. It is beyond the limits of what they can think. The only thing we can get is the resent of the community.

This paper tries to "solve" problems by adding layers of abstraction. It is never going to work. The only thing that will work is to go back

an fix the errors made.

The other link has much lesser pretensions to the point I fail to see what is their contribution. Is the formalization of compartmental modeling something needed? Specially, which is the point of introducing mathematical precision into it and finally writing an ODE jumping over any rationality as we have been doing for years?

The relation between tools and problems is dialectic. Newton's physics called for differential calculus and differential calculus made possible Newton's physics. One calls for the other. Poincaré's topology was needed trying to solve the three-body problem, it did not pre-existed the problem. Electromagnetism uses and abuses Gauss' formula and Helmholtz theorem. They where created for it. But this dialectic is broken by specialization. Hence it ends at the beginning of the XX century in physics.

If we teach to use tools and do not show how the construction of the tools was dictated by the problem and helped to specify the problem, and then they form a duality what we are doing is to create professionals that shall use instructed ideas (other people ideas) to confront "problems". We have predefined cages and force nature into cages with as much force as it is needed. This is how the professional scientist is formed and the form we reproduce ourselves. Unfortunately, this process is incompatible with critical thinking. We have been impaired and in turn we impair our students of producing critical thinking. I live under the impression that today's scientists do not really know what it is about.

I came to this moral dilemma some years ago: If I accept to guide or counsel a student in his studies I have to help him/her to achieve what I consider is the best development of a human being. But if such development is going to make her a social pariah, which right do I have to ruin his life?

I then decided not to ruin the life of any student and accepted no more students.

I will ask around to see if there is somebody that could match your interests and let you know.

In the mean time you may want to check this researcher:

CHRISTIAN DE RONDE

https://www.conicet.gov.ar/new_scp/detalle.php?keywords=CHRISTIAN%2BRONDE&id=38053

who has worked on the logos categorical approach. I do not know him, his key words are: quantum entanglement, quantum mechanics, relationalism, category theory, potentia.

It is not a recommendation, you have to check him out.

I will get back to you if I get better information.

All the best

Hernán

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> learning more about Category Theory, in particular Topos Theory,
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> <<https://topos.site/blog/2023/05/categories-for-epidemiology/>>,

[Texto das mensagens anteriores oculto]

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Hernán G Solari

<http://users.df.uba.ar/solari/>

Hernán Gustavo Solari <hgsolari@gmail.com>
Para: Nicholas Funari Voltani <nicholasvoltani@gmail.com>

7 de novembro de 2023 às 18:33

Dear Nicholas

I asked Federik Holik about Christian De Ronde. Federico is a young scientist in the philosophy of science that was my student during his undergrad years and later developed his PhD with Olimpia Lombardi, the highest reference for the philosophy of science in Argentina (check her at <https://plato.stanford.edu/search/searcher.py?query=olimpia+lombardi>)
Olimpia just retired, but I appreciate Federico very much, even when we disagree.

Federico worked with "de Ronde" some papers and says that Christian de Ronde is recommendable if what you intend is to be an academic in philosophy. Instead, if what is in your interest is more towards mathematics or physics and a philosophical view of them, he would not recommend him. I have reviewed one project by Christian de Ronde and have the same impression.

The good thing is that Federico Holik <olentiev2@gmail.com> works in logical and mathematical aspects of quantum mechanics and is familiar with "Topos". Perhaps his approach is of interest to you. You can write to him without compromise, he is a great person and will help you to make clear your ideas.

With respect to the Heidegger in the paper you pointed to, Federico says is just decoration. It can be suppressed. From a technical point of view that line of research appeared first as promising but later showed technical problems. Federico stopped tracking their work.

all the best

Hernán

[Texto das mensagens anteriores oculto]

Nicholas Funari Voltani <nicholasvoltani@gmail.com>
Para: Hernán Gustavo Solari <hgsolari@gmail.com>

9 de novembro de 2023 às 22:03

Hello professor. Thank you for the responses, and sorry for the delay, too much work and all! I'll get in touch with Federico to get some information on his field of work, it seems interesting.

Regarding your first response, the first thing is: do you write somewhere, as in a blog? It's clear that these topics are really important to you, and you write very elegantly about them. I'd really get to read more, aside from papers that you published.

I have some thoughts about what you wrote, please bear with me:

You brought up the notion that Science is becoming more and more specialized, atomized (due to capitalism and all), which goes against the "whole-ness" of Nature, which Itself doesn't "partition" itself into subjects— it just *is* (which brings to mind a Taoist view of Nature, which I share). I feel that knowledge — which has to be *human* knowledge, written in humanly-comprehensible categories, if it be of any interest to us — ends up being "partitioning", inevitably, due to our finite nature (both physically and, consequently, psychologically and rationally). This already brings the question: what, then, is this limit of reason? We don't have to be as specialized as academia is pushing Science to be, but how far can we really go as far as "scientific holism" is concerned, while maintaining a "credible" theory, able to make predictions, etc?

In a similar theme, you also bring up the dialectic relation between "tools" and "problems", which I find very interesting and can see that it makes sense... but do both *always* have to be related dialectically? I understand it's ideal for this relation to be dialectical, "organic" so to say, which is to say that, if this relation isn't so, then it is "contingent", possibly contradictory (in the bad sense of the word, not in the to-be-expected hegelian contradictions)... perhaps it's the way in which I was brought up, because it feels that it'd put so many theories and things to waste, things which even work, and may even work well in some cases... In the same token, I agree wholeheartedly in the sense that, to overcome a theory, one has to bypass its old ways and "create itself anew", so to say — which sounds a lot like Hegel's *Aufhebung* (from the little that I know).

I understand your point on critical thinking, in the sense that the "way out" of this sea of models and contingent theories has to be through Reason, albeit with "intellectual honesty", acknowledging explicitly one's hypotheses to be best of one's ability. Since I see you were somewhat brutally honest with me, I feel I can be at least somewhat candid as well. The truth is that I'm currently not sure which way I want to follow in research. I don't feel 100% attracted to Physics nor pure Mathematics; I've always felt affinity with the Humanities, but I don't know whether I'd like working purely within it; I am interested in a combination (a superposition?) of these themes. Recently I've enrolled in an Epistemology/Philosophy of Physics course in USP, and went as I could; although the course was too shallow, it's something that piques my interest. There's a phrase that caught my attention in a text that you co-published, which says that "*Estamos persuadidos que la razón se da naturalmente en el ser humano y brota por sí sola. Desplazada del sistema académico en buena medida, crecen allí,] y en muchos otros intersticios, nuevos retoños. En esos brotes se cifra la esperanza de la civilización de superar la crisis*". I understand what you say in your response, that this kind of intellectual work, this true "critical thinking" which you mention, isn't really what makes one's academic career take off, but, honestly, there are things which satiate the body and things which satiate the soul, and these discussions and ideas, which I see that clearly occupy your waking hours, are things that I seek as well, on a personal level: to disentangle ideas which have become too entangled to be correctly ascertained; to make clear sense of what is ill-defined and ill-discussed; if not to pursue Truth, then to dispel what is untrue, because, while there may not be a capital-T Truth, there certainly are many things which are false, some of which can be extremely dangerous not to be undone.

I appreciate the care and honesty that you shared, but I honestly don't think that getting to learn to be more critical of the world around us is as drastic a fate as you put it. Also, if all goes wrong in academia for me, I can safely return to the corporate world and pay my bills (which I wouldn't prefer, but it's a safeguard, due to my CV). So, if you'd have the patience, and if you find me somewhat "worthy" of it, I'd really like to learn more with you, not necessarily in a research scenario, but perhaps in some sort of "mentoring" fashion.

In any case, I'd like to know where I can learn more from what you've spoken. I reckon references like Hegel's "Science of Logic" would be a bit too much for me as of now, so I'd accept some simpler recommendations (can be in English or even in Spanish, yo sólo no he empezado todo en español porque puedo expresarme mejor en inglés, entonces preferí hacerlo, pero también puedes responderme en Español si lo prefieras). Thanks!

[Texto das mensagens anteriores oculto]

Hernán Gustavo Solari <hgsolari@gmail.com>
Para: Nicholas Funari Voltani <nicholasvoltani@gmail.com>

10 de novembro de 2023 às 12:24

Dear Nicholas

thank you for your letter, I truly appreciate your intellectual honesty and the trust in me that your letter reflects.

I have a 30 yo daughter that is studying Atmospheric Science (PhD). The question whether Academia is the best place to be, a good one or not even good, is a debate of every day. She was first lured by molecular biology. After one and a half years of studying with perfect grades she understood that she was being prepared to be a sort of human-machine, a high functioning human machine, a flesh-robot that thinks with implanted thoughts not her own thoughts, not her own values. She had to learn how to do and no body cared whether she understood, much less approved what she was doing. The mass society tells us that if we resign our freedom we will be "granted" access to the material goods. The motto would be something like: let your materiality blossom. But it goes without saying that to become mass-people you have to kill the unique being that you call by your name or simply "I". How high functioning silica-robots are going to change this? I do not know, but there is an observable convergence between mass-people and robots, between flesh-robots and silica-robots. The world of AI wants us to believe that silica-robots "think" the way we do. I am afraid that the only thing their believe proves is that AI-scientists are flesh-robots. But, can the AI-scientist believe otherwise? Yang Tze's sayings in "Autumn floods" tells that it is almost impossible (see below).

How do we have to help people develop their "bildung"? (a term from Wilhelm von Humboldt that roughly means "self construction" and was the center of Humboldt's ideal of university). Certainly not by stuffing them with information, or training them at performing certain tasks. Humboldt said of such things "we admire what they do but despise what they are". (W von Humboldt, The limits of action of the state).

Have you read "La rebelión de las masas" by Ortega y Gasset? It was written in 20's in Spain and discusses the contrast between the mass-society and the Enlightenment society. Each chapter appeared as a note in a newspaper. Chapter 10 or 11 is about the scientists as the highest form of the mass men. The chapter's title: la catástrofe del especialismo. In that chapter you will find your own words regarding our approach to knowledge as specialists and the impossibility for nature to be partitioned in realms. About the same time, José Ingenieros (Argentina) wrote "El hombre mediocre" shaken by the same kind of observations. It was the time of ascent of the fascism here and there. Together with the shorter writing by Kant: *Qué es la ilustración?* (also appeared in a Newspaper but by ~1780) would give you an idea about the abyss between the "Sapere aude" of the autonomous person and the more common heteronomous strategy of imitation, doing as others do.

If we conceive understanding as accumulation of information/facts. The extension of the field suggests that we have not other possibility than becoming an specialist, somebody that knows a lot about a small part (una porción) says OyG) of the Universe and ignores everything about the rest of the universe. Let me copy+paste OyG

"He aquí un precioso ejemplar de este extraño hombre nuevo que he intentado, por una y otra de sus vertientes y haces, definir. He dicho que era una configuración humana sin par en toda la historia. El especialista nos sirve para concretar enérgicamente la especie y hacernos ver todo el radicalismo de su novedad. Porque antes los hombres podían dividirse, sencillamente, en sabios e ignorantes, en más o menos sabios y más o menos ignorantes. Pero el especialista no puede ser subsumido bajo ninguna de esas dos categorías. No es un sabio, porque ignora formalmente cuanto no entra en su especialidad; pero tampoco es un ignorante, porque es aún hombre de ciencia. y conoce muy bien su porción de universo! Habremos de decir -que es un sabio ignorante, cosa sobremanera grave, pues significa que es un señor el cual se comportará en todas las cuestiones que ignora, no como un ignorante, sino con toda la petulancia de quien en su cuestión especial es un sabio."

But there is a different concept of understanding, more human, rooted in our childhood, to understand would be to give coherence and unity to what is observed. To provide a logical structure that makes it thinkable and therefore predictable up to a certain point. This action happens (it would happen if you prefer) under a disposition that we share at least with mammals: reasoning. The rational person does not know of everything, she is not an Encyclopedia, she is only in command of an innate method (whose rules we barely intuit but deserve to be written). With the method and provided with observations and some time she can construct a logical structure for the observations and phenomena. The wise woman does not have the answers to every thing, but she can seek them with economy of effort. In so doing she will realize that there is a unity in (her perception) of the world and that such a unity is a reflection of her inner unity. The unity of reason and the unity of ourselves. And finally, the understanding that there is just one LIFE (capital letters) in which our life partakes. In Hegel (an idealist) it would be "THE IDEA". In as much and to the extent that physics is rationally constructed, she will be a physicist speaking of physics, although she would have reservations at some points. Speaking of biology she would be a biologist. Speaking about Justice she will know the fundamentals, as a supreme court judge, but will mostly ignore the procedural trickery that abounds in ordinary justice. She would guess the possible existence of a law saying xxxx but ignore the actual existence and the

details until she finds it.

Reasoning is an art, it is not just logic, it contains logic in it but cannot be reduced to it. And reasoning is self improving.

About dialectics. For most people that have being brainwashed by our school system, access to dialectics is closed. I know great academics in philosophy that very honestly confess to us that they cannot understand dialectic and ask for our help. Unfortunately, we cannot help, dialectics comes from within. It is not a procedure, it is a form in which we seek harmony with the world, has more to do with the Tao, than with logic. If you have affinity with dialectic thought, reading Hegel's logic could go slow but it is a good place. My need for Hegel's logic came at age 16 or 17 and has been a companion all my life. Did I understand something at such early age? Very little, and very little I understand today, but what it says make me think, it resonates in my brain. We do not aim at being commentators of Hegel, the critic of art to whom you cannot trust a brush. We want to be artists.

In the same form, the academic in philosophy does not necessarily is a philosopher, most of times he is not. According to Nisbet ("The degradation of the academic dogma") there are two kind of academics. The scholastic style is always present at universities. The work within a school of thoughts which is a social community as well. He works by associating his thoughts to the thoughts of the community's authority. The other kind is not always present, his thoughts are directed straight to Nature, their dialogue is with Nature.

We read in a book I have yet to read (A kinship with all life, by John Allen Boone,
<https://archive.org/details/kinship-with-all-life-john-allen-boone>)

"There's facts about dogs, and there's opinions about them. The dogs have the facts, and the humans have the opinions. If you want facts about a dog, always get them straight from the dog. If you want opinions, get them from the human"

With regard to Federico Holik. I believe his work is good in terms of academic philosophy of science, he is unhappy with the current academic system and he believes we can/must develop our own projects in parallel. He has not seen the tao so far. My favorite quote of Zhang Tze ends: 'Jo of the North Sea said "... Now you have come out beyond your banks and borders and have seen the great sea —so you realize your own pettiness. From now on it will be possible to talk to you about the Great Principle."

Another friend is a Sociologist quite fond of Edgar Morin and "complex thinking". Leonardo Rodriguez Zoya <leonardo.rzoya@gmail.com> among other initiatives keeps a site: <https://pensamientocomplejo.org/> with lots of material, specially from South and Central America. He is also somebody interesting to know.

All of us used to be part of meetings at Alejandro Romero's place. Alejandro is at the margins of the academic system but he is most admired by us. He used to held meetings in his home where young people (and not so young) discussed their views and tried to address some topic of common interest writing essays to be discussed in the group and, may be, later published. <romerofernandezalejandroguille@gmail.com> (In case you write to him do it in Spanish, perhaps Portuguese works, but not English).

All this activity is "not for profit", we simply share our thoughts because we enjoy thinking and challenges. You are welcome to ask as many questions as you want and to comment whatever you find of interest. If we are not going to ruin your academic life, things are easier.

All the best
 Hernán

PD: I may have forgotten answering something. If that is the case, simply remind it to me. I do not have a blog.

> Hello professor. Thank you for the responses, and sorry for the
 > delay, too much work and all! I'll get in touch with Federico to get
 > some information on his field of work, it seems interesting.
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 > Nature, which I share). I feel that knowledge — which has to be
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 > medida, crecen allí[,] y en muchos otros intersticios, nuevos
 > retoños. En esos brotes se cifra la esperanza de la civilización de
 > superar la crisis*"*. I understand what you say in your response, that

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